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Experiment No : 04

Experiment Name : Measurement of e/m of an electron using an electron beam deflection tube

Theory:

When current is passed through the filament, it becomes hot and electrons gain sufficient energy to escape from the filament. This process is known as thermionic emission. When a charged particle moves through a region in which both electric and magnetic fields are present, it experiences a force given by the Lorentz force law,

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

If the charged particle is an electron of charge e moving in a magnetic field and the electric field is absent, then,

$$\vec{F} = e(\vec{v} \times \vec{B})$$

$$\Rightarrow F = evB \sin \theta$$

$$\Rightarrow F = evB \quad [\theta = 90^\circ]$$

The magnetic force provides the necessary centripetal force for circular motion, so...

$$\frac{mv^2}{r} = evB$$

$$\Rightarrow v = eBr/m$$

If the electron is initially accelerated by an applied potential difference V_a , then we have,

$$eV_a = \frac{1}{2}mv^2$$

$$\Rightarrow \frac{e}{m} = \frac{2V_a}{B^2 r^2}$$

The magnetic field is produced by a pair of Helmholtz coils, which generates a nearly uniform magnetic field near the center. The magnitude of the field is given by,

$$B = \mu_0 \left(\frac{4}{5} \right)^{3/2} \frac{nI}{a} = 8.99 \times 10^{-7} \frac{nI}{a}$$

Where,

n = number of turns in each of the pair of coils

I = current in the coil

a = radius of the coil

μ_0 = permeability of the empty space

$$= 4\pi \times 10^{-7} \frac{\text{wb}}{\text{Amp-m}}$$

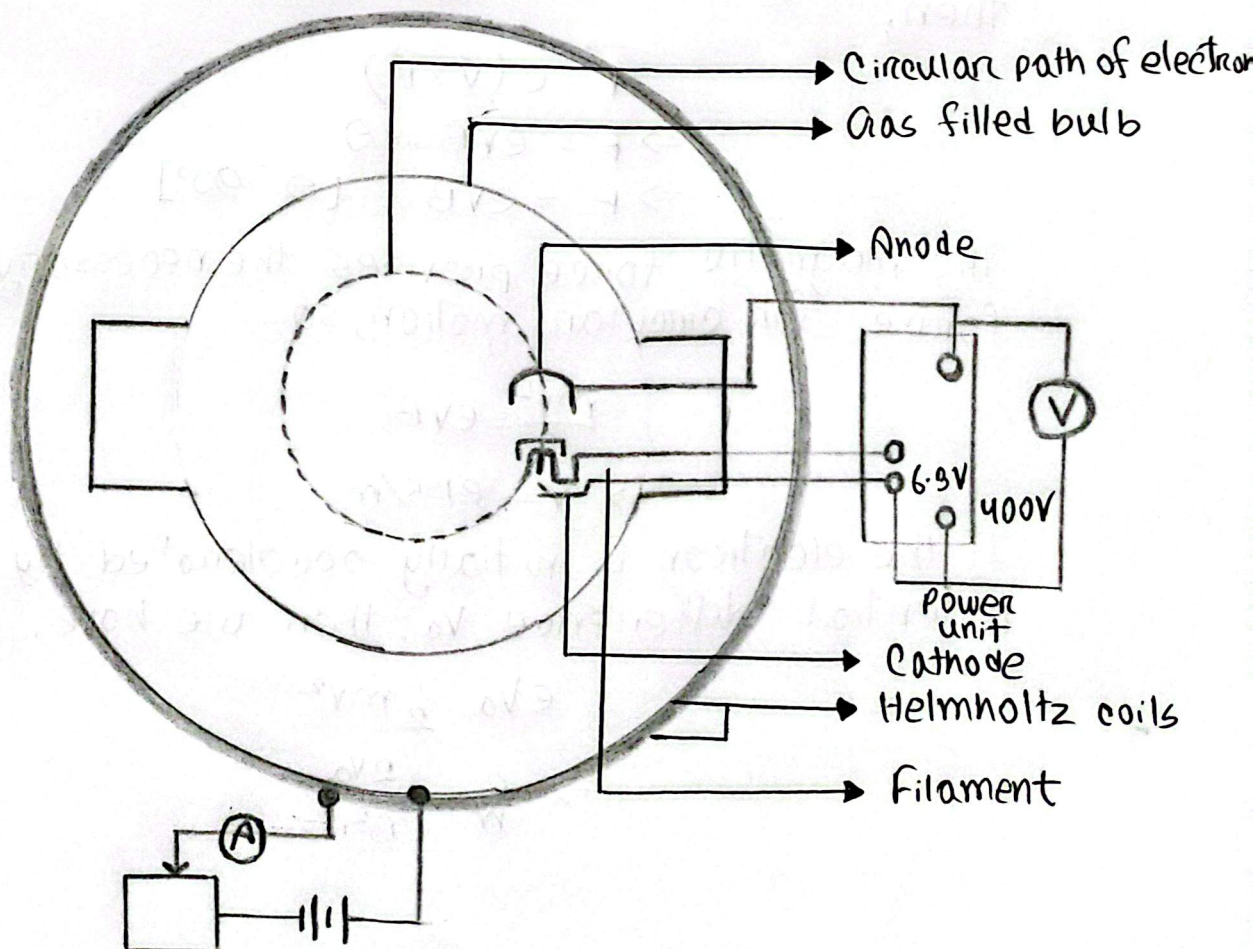


Figure 1: Electron Beam Deflection Tube

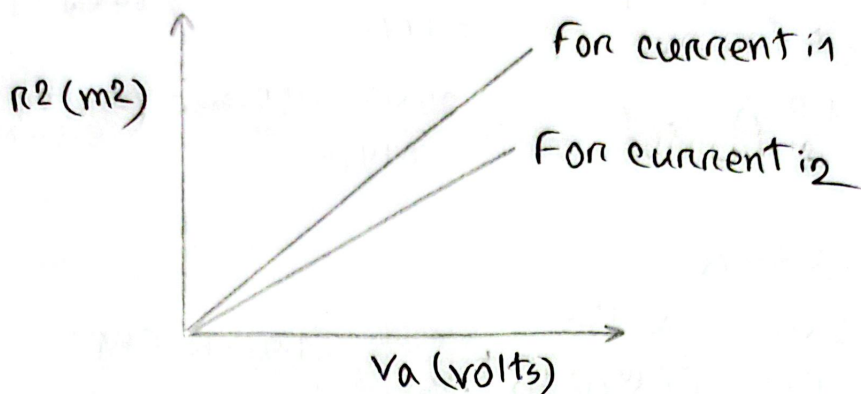


Figure 2: r^2 Vs V_a graph

Required Apparatus:

1. Electron beam deflection tube with Helmholtz coils
2. Power supply

Calculations:

From graph,

$$m_1 = \frac{AB}{BC} = \frac{0.52 \times 10^{-3}}{31.5} = 0.0165 \times 10^{-3}$$

$$\Rightarrow (e/m)_1 = \frac{2}{B^2} \times \frac{1}{m_1} = 2.08 \times 10^{11} \text{ C/kg}$$

~~from~~

$$m_2 = \frac{DE}{EF} = \frac{0.76 \times 10^{-3}}{73.5} = 0.0103 \times 10^{-3}$$

$$\Rightarrow (e/m)_2 = \frac{2}{B_2^2} \times \frac{1}{m_2} = 2.077 \times 10^{11} \text{ C/kg}$$

$$(e/m)_{\text{avg}} = \frac{(e/m)_1 + (e/m)_2}{2} = 2.079 \times 10^{11} \text{ C/kg}$$

$$\% \text{ Error} = \frac{|1.75 \times 10^{11} - 2.079 \times 10^{11}|}{2.079 \times 10^{11}} \times 100 = 15.8 \%$$

For, $I = 0.95 \text{ A}$, $B = \frac{8.99 \times 10^{-7} \times 132 \times 0.95}{0.1475} = 7.64 \times 10^{-4} \text{ T}$ $[B = 8.99 \times 10^{-7} \times \frac{nI}{a}]$

For, $I = 1.2 \text{ A}$, $B = \frac{8.99 \times 10^{-7} \times 132 \times 1.2}{0.1475} = 9.65 \times 10^{-4} \text{ T}$

For, $I = 0.95 \text{ A}$

(i) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 175}{(7.64 \times 10^{-4})^2 \times 3.025 \times 10^{-3}} = 1.98 \times 10^{11} \text{ C/kg}$

(ii) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 148}{(7.64 \times 10^{-4})^2 \times 2.5 \times 10^{-3}} = 2.028 \times 10^{11} \text{ C/kg}^{-1}$

(iii) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 128}{(7.64 \times 10^{-4})^2 \times 2.025 \times 10^{-3}} = 2.166 \times 10^{11} \text{ C/kg}$

(iv) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 113}{(7.64 \times 10^{-4})^2 \times 1.6 \times 10^{-3}} = 2.42 \times 10^{11} \text{ C/kg}$

(v) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 105}{(7.64 \times 10^{-4})^2 \times 1.225 \times 10^{-3}} = 2.94 \times 10^{11} \text{ C/kg}$

For, $I = 1.2 \text{ A}$

(i) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 241}{(9.65 \times 10^{-4})^2 \times 2.76 \times 10^{-3}} = 1.88 \times 10^{11} \text{ C/kg}$

(ii) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 200}{(9.65 \times 10^{-4})^2 \times 2.26 \times 10^{-3}} = 1.9 \times 10^{11} \text{ C/kg}$

(iii) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 173}{(9.65 \times 10^{-4})^2 \times 1.8 \times 10^{-3}} = 2.06 \times 10^{11} \text{ C/kg}$

(iv) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 139}{(9.65 \times 10^{-4})^2 \times 1.4 \times 10^{-3}} = 2.13 \times 10^{11} \text{ C/kg}$

(v) $\frac{e}{m} = \frac{2V_0}{B^2 n^2} = \frac{2 \times 116}{(9.65 \times 10^{-4})^2 \times 1.06 \times 10^{-3}} = 2.35 \times 10^{11} \text{ C/kg}$

from calculation,

$$\left(\frac{e}{m}\right)_{\text{avg}} = \frac{1.98 + 2.028 + 2.166 + 2.42 + 2.94 + 1.88 + 1.9 + 2.06 + 2.13 + 2.35}{10} \times 10^{11}$$

$$= 2.1854 \times 10^{11} \text{ C/kg}$$

$$\therefore \text{Error} = \frac{|1.76^{88} \times 10^{11} - 2.185 \times 10^{11}|}{1.76^{88} \times 10^{11}} \times 100\%$$

$$= \frac{24.25}{24.26} \%$$

Experimental Data:

Number of turns in each of the pair of coils, $n = 132$ ✓

Coil radius, $a = 0.1475 \text{ m}$ ✓

Table 1 - Data for determination of $\frac{e}{m}$ for different coil current

Coil current I (A)	Magnetic field B (T)	Accelerating Voltage V_a (V)	Diameter d (m)	radius $r = d/2$ (m)	r^2 (m ²)	$\frac{e}{m}$ (C/kg)
0.95	7.64×10^{-9}	175	0.11	0.055	3.025×10^{-3}	1.98×10^{11}
		148	0.10	0.05	2.5×10^{-3}	2.028×10^{11}
		128	0.09	0.045	2.025×10^{-3}	2.166×10^{11}
		113	0.08	0.04	1.6×10^{-3}	2.42×10^{11}
		105	0.07	0.035	1.225×10^{-3}	2.94×10^{11}
1.20	9.65×10^{-9}	241	0.105	0.0525	2.76×10^{-3}	1.98×10^{11}
		200	0.095	0.0475	2.26×10^{-3}	1.98×10^{11}
		173	0.085	0.0425	1.8×10^{-3}	2.06×10^{11}
		139	0.075	0.0375	1.4×10^{-3}	2.13×10^{11}
		118	0.065	0.0325	1.06×10^{-3}	2.935×10^{11}

Results:

$$\frac{e}{m} = 2.1854 \times 10^{11} \text{ C/kg}$$

$$\frac{e}{m} = 2.079 \times 10^{11} \text{ C/kg}$$

$$\% \text{ of error} = 24.26$$

$$\% \text{ of error} = 18.8$$

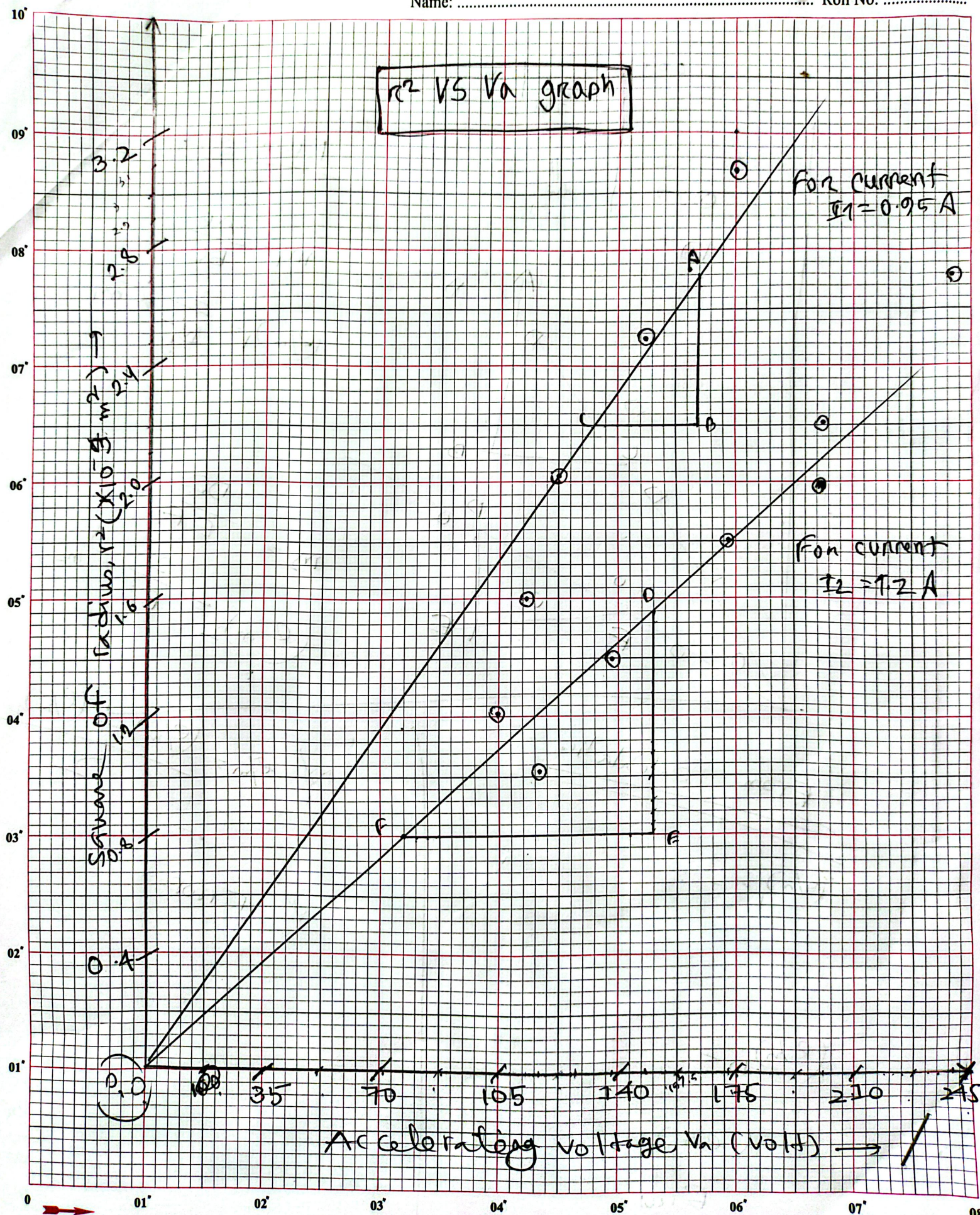
... (from calculation)

... (from graph)

... (from calculation)

... (from graph)

18.01.20



Precautions:

- Care was taken to determine B as accurately as possible
- The path of the electrons was made as circular as possible
- Parallax error was avoided while taking reading for the diameter
- If the anode voltage is too high, tube lifetime will be shortened. The voltage was not raised higher than necessary. Usually a few hundred volts (max 275 V) was enough